Approximation and Representation

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1 Numbers

A *geometric sequence* $a_1, a_2, ...$ is one where each successive term in the sequence is obtained via multiplication by a common factor on the most recent term. For example, each of the following is a geometric sequence:

$$2,\ 4,\ 8,\ 16,\ 32,\ ... \tag{1.A1}$$

$$1,\ \tfrac{2}{3},\ \tfrac{4}{9},\ \tfrac{8}{27},\ \tfrac{16}{81},\ ... \tag{1.A2}$$

$$a,\ ar,\ ar^2,\ ar^3,\ ... \tag{1.A3}$$

The general form of a geometric sequence given in the lattermost representation says that, if we start with the first term $a_1 \equiv a$, then each successive term $a_2, a_3, ...$ is obtained via multiplication by the common factor r, which we call the *common ratio* of the sequence $\{a_n\}_{n=1}^{\infty}$. (We mention the notation: in general, we use a_k with the index $k \in \mathbb{N}$ taking values in the positive integers to represent the kth term in the sequence. Thus, a sequence $a_1, a_2, ...$ may be expressed by, for example: $\{a_n\}_{n=1}^{\infty}$, $\{a_k\}_{k\in\mathbb{N}}$, $\{a_1, a_2, ...\}$, and so on. As we continue, we may even simply write a_n with the understanding that the index n typically means "an index that varies over the natural numbers $\mathbb{N}$." None of these notations should be entirely surprising, and we use each freely.)

There is a clear distinction between a sequence like (1.A1), where the terms get larger and larger as we progress, compared to one like (1.A2) whose terms get smaller and smaller. In the notation of (1.A3), we observe that sequence (1.A1) has $a = 2$ with $r = 2$, while (1.A2) has $a = 1$ and $r = 2/3$ (check this!). It is immediately clear that the growth in (1.A1) is a result of the multiplication by $r = 2 > 1$, while the shrinkage in (1.A2) results from the multiplication by $r = 2/3 < 1$.

Sequences on their are of major interest when studying algorithmic procedures — for instance, if we claim that some numeric optimization method produces smaller and smaller error as we continue to iterate, we may want to test the performance of our algorithm against known test-cases whose analytic results are known. In such a case, we may call the error at the nth iteration of the algorithm $\epsilon_n = |a - a_n|$, where a_n would represent the numeric result produced by our algorithm at the nth iteration, while a is the analytic result to which we hope our algorithm *converges* as we continue to iterate. If we are confident that *eventually* it will be the case that the algorithm converges to the value a, we might say that a is the *limit* of the sequence a_n, and write

$$a := \lim_{n\to\infty} a_n. \tag{1.A4}$$

Relative to our $\epsilon_n = |a - a_n|$, it would follow that whenever (1.A4) holds, we would have $\epsilon_n \to 0$ as $n \to \infty$. (Note the two notations: if we were to call $\epsilon := 0$, then we could write $\lim_{n\to\infty} \epsilon_n = \epsilon$.

It can often feel a bit more natural to write "the sequence $\epsilon_n \to \epsilon$ as $n \to \infty$," which omits the lim symbol explicitly but remains implicit in the description. We may also write $a_n \overset{n\to\infty}{\longrightarrow} a$ to denote the convergence of the sequence a_n to the limit a as the index $n \to \infty$).

On their own, sequences are useful bookkeeping devices. However, beyond the growth rate as we move down the line of a sequence, a major interest often lies in understanding its *collective behavior*. For instance, we understand from (1.A1) that the individual terms in the sequence $a_n \to \infty$ as $n \to \infty$. It may not be entirely clear how we could formalize this notion; to do so, we can make use of the following definition, which also formalizes some of the ideas from above.

Definition 1.0.1 (Sequential Convergence)

Let $\{a_k\}_{k\in\mathbb{N}}$ be a sequence. We say that a_k *converges to the limit* a if *for any* arbitrarily small $\epsilon > 0$, we can find a thresholding index $N \in \mathbb{N}$ such that for any $n \geq N$ indexing a member a_n at least as deep in the tail of the sequence as a_N, we have

$$|a_n - a| < \epsilon.$$

That is, say that a_n converges to the limiting point a if for any $\epsilon > 0$,

$$\text{There exists } N \in \mathbb{N}\colon \quad \text{For all } n \geq N,\ |a_n - a| < \epsilon. \tag{1.A5}$$

If no such limiting point a exists (i.e., if for any proposed limit a and any threshold N, we can find an index $n \geq N$ such that the distance $|a_n - a| \geq \epsilon$ *for some* $\epsilon > 0$), we say that the sequence a_n *fails to converge* (or that a_n is *nonconvergent*). If $a_n \to \pm\infty$ as $n \to \infty$, we say that a_n *diverges* (to $\pm\infty$, as appropriate).

The failure of a sequence $\{a_k\}_{k\in\mathbb{N}}$ to converge is loosely the property that for any proposed limit a, we can always find an $\epsilon > 0$ such that a_n fails to be ϵ-close to a at some point in the tail of the sequence. There are many types of failure we can imagine. Notice the subtle difference in the characterizations: a sequence converges if and only if for *any* $\epsilon > 0$, there exists $N \in \mathbb{N}$ such that $|a_n - a| < \epsilon$ for all $n \geq N$. On the other hand, to establish the converse, we only need to identify *some* $\epsilon > 0$ such that for any fixed index $N \in \mathbb{N}$, $|a_n - a| \geq \epsilon$ for *at least one* $n \geq N$. The point is that while a sequence which fails to converge can be quite easy to establish, convergence can be a bit trickier; the former needs us establish only one counterexample, while the latter needs a guarantee that holds for any $\epsilon > 0$.

Nevertheless, establishing the failure of a sequence to converge can be made simpler. In our characterization above, we said that a_n is a nonconvergent sequence if no limit point a exists. But as it stands, this would clearly require us to check every single limiting point a against our criterion; for instance, it seems somewhat clear that the sequence (1.A1) is nonconvergent, since the terms $2, 4, 8, 16, 32, \ldots$ continue to get bigger and bigger. Intuitively, there is no way we could pick a single real number $a \in \mathbb{R}$ and find a member of the sequence $a_N \in \{2, 4, 8, 16, 32, \ldots\}$ such that $|a_n - a| < \epsilon$ for all $n \geq N$ if we were to take, say, $\epsilon = 1$. It would thus seem immediately useful to establish a criterion that would allow us to characterize whether a sequence converges or not independently of arbitrarily proposed limit points a. One way to do this is to notice that, if we look at *pairs of successive entries* (a_1, a_2), (a_2, a_3), and so on, the *pairwise distances* grow larger and larger as time goes on — in particular, $|a_2 - a_1| = |4 - 2| = 2$, $|a_3 - a_2| = 4$, $|a_4 - a_3| = 8$, and so on.

While a useful place to start, we must take some care in establishing this heuristic, because as stated it is too loose for some sequences whose terms exhibit a deceptively slow rate of decay. That is, even though

$$a_n \overset{n\to\infty}{\longrightarrow} a \implies |a_{n+1} - a_n| = |a_{n+1} - a + a - a_n| \leq |a_{n+1} - a| + |a_n - a| \overset{n\to\infty}{\longrightarrow} 0,$$

the converse *fails*. For instance, if we take[1]

$$a_n = \log n,$$

then

$$|a_{n+1} - a_n| = |\log(n+1) - \log(n)| = \left|\log\left(\frac{n+1}{n}\right)\right| = \left|\log\left(1 + \frac{1}{n}\right)\right| \overset{n\to\infty}{\longrightarrow} \log(1) = 0.$$

This is clear simply because the sequence $\{1/n\}_{n\in\mathbb{N}}$ has terms $1, 1/2, 1/3, 1/4, \ldots$ converging to 0 as $n \to \infty$. In particular, given $\epsilon > 0$, if we propose the limit $a = 0$ with $a_n = 1/n$, then

$$|a_n - a| = \left|\frac{1}{n} - 0\right| = 1/n \overset{set}{<} \epsilon \implies n > 1/\epsilon,$$

so that in order to get $a_n = 1/n$ within ϵ of the limit $a = 0$, we need simply look at the terms in the tail a_n for $n > 1/\epsilon$; any such term will satisfy the convergence definition (1.A5) relative to the given $\epsilon > 0$. This shows that, indeed, $a_n = 1/n \to 0$ as $n \to \infty$. It is thus certainly the case that

$$b_n := 1 + \frac{1}{n} \overset{n\to\infty}{\longrightarrow} 1,$$

since clearly the addition of a constant 1 to the convergent sequence $1/n$ doesn't affect the convergence of $1 + 1/n$, and

$$c_n = \log\left(1 + \frac{1}{n}\right) \overset{n\to\infty}{\longrightarrow} \log(1) = 0,$$

since taking the logarithm of numbers slightly larger than and consistently decaying toward 1 will result in logarithms which are slightly larger than and continuously decaying toward 0. Nevertheless, what we *also* know is that the natural logarithm is an *unbounded function*, in the sense that

$$\lim_{n\to\infty} \log(n) = \infty.$$

What this means is that if we look at the successive values of the original sequence $a_n = \log n$, we'll observe that $a_n \to \infty$ *even though* the pairwise distances $|a_{n+1} - a_n| \to 0$ as $n \to \infty$. All of this is to convince you that it is not sufficient to merely consider whether *pairwise distances* get small

[1]Throughout, the symbol log is used to denote the natural logarithm of base e — i.e., we use log instead of ln. This is largely because the base of the logarithm plays a less substantial role in limiting scenarios, as is apparent from the change of base formula — see Exericse XX.

and remain small in order to establish whether or not a sequence a_n converges to *some* finite limit. (Again, the interest at this point is *only* in "whether or not the sequence converges"; the question of "what it converges to" is the topic of future discussion).

All of this may cast doubt on the pursuit of a limit-agnostic criterion constructed relative only to properties of the sequence a_n itself. The trick, however, is that we should not ask whether *adjacent* terms eventually become close. We've observed that convergence is stronger than that: if a convergent sequence $a_n \to a$, then we imagine we can choose a point in the tail of the sequence a_N such that the distances $\epsilon_n := |a_n - a|$ get small and stay small for any $n \geq N$. Indeed, what's clear is that if $a_n \to a$, then there exists an N such that $|a_n - a| < \epsilon$ for any $\epsilon > 0$. But since this has to hold for *any* such $\epsilon > 0$, we can just as well force it to hold for $\epsilon' := \epsilon/2$ by finding an appropriate index N'. In this case, the triangle inequality implies that for any two indices $n, m \geq N'$, each of $|a_n - a| < \epsilon' \equiv \epsilon/2$ and $|a_m - a| < \epsilon'\epsilon/2$, so that

$$\text{For all } n, m \geq N' : \quad |a_n - a_m| = |a_n - a - (a_m - a)| \leq |a_n - a| + |a_m - a| < 2\epsilon' \equiv \epsilon.$$

To summarize, what we've just established is that if $a_n \to a$, then for any $\epsilon > 0$, we can identify a thresholding index $N \in \mathbb{N}$ such that the distance between *any two* points further in the tail a_n and a_m, $n, m \geq N$ gets small and stays small:

$$a_n \overset{n\to\infty}{\longrightarrow} a \implies \text{Given any } \epsilon > 0 : \Big(\text{There exists } N \in \mathbb{N} : \quad \text{For all } n, m \geq N, \quad |a_n - a_m| < \epsilon\Big). \tag{1.A6}$$

We now state the formal characterization of this phenomenon.

Definition 1.0.2 (Cauchy Sequences)

A sequence $\{a_n\}_{n\in\mathbb{N}}$ is said to be a *Cauchy sequence* if for any $\epsilon > 0$, there exists a threshold $N \in \mathbb{N}$ such that for all $n, m \geq N$,

$$|a_n - a_m| < \epsilon.$$

Intuitively, we can imagine that a Cauchy sequence $\{a_n\}_{n\in\mathbb{N}}$ is one where no matter how small the window $\epsilon > 0$, we can always find a point in the tail of the sequence such that everything beyond fits inside an interval of length ϵ. This immediately clarifies why our previous attempt failed to establish a useful criterion: for a sequence like $c_n = \log(n)$, though it is the case that $|\log(n+1) - \log(n)| \to 0$ as $n \to \infty$, it is not the case that $|\log(m) - \log(n)| \to 0$ for a fixed n if we're allowed to continue varying m. Concretely, if we take $\epsilon = 1$, then for any proposed thresholding index $N \in \mathbb{N}$, choose $n \geq N$, and then choose $m \geq n$ so large that $m \geq en$ (again, this is possible because $\log(n) \to \infty$ as $n \to \infty$). In this case, we have

$$|\log(m) - \log(n)| = \left|\log\left(\frac{m}{n}\right)\right| \geq \left|\log\left(\frac{en}{n}\right)\right| = \log(e) = 1 \equiv \epsilon.$$

Thus, no matter how deep in the tail we go, we can always find two terms whose distance is at least 1.

At this point, we've established the useful criterion that if a sequence a_n converges to a limit a, then the sequence is Cauchy: for any $\epsilon > 0$, there exists a threshold $N \in \mathbb{N}$ such that for any $m, n \geq N$,

$$|a_n - a_m| < \epsilon.$$

Really though, we've yet to establish the desired converse: what we'd like is to be able to say that if a sequence a_n is Cauchy, then it converges to a limit a. The utility in the converse is that whether a sequence is Cauchy can be ascertained completely from the *construction of the sequence itself* and, importantly, ambivalent to a proposal limit a. If we don't know whether a sequence converges, then the matter of proposing a limit to check its convergence is only productive if we have some intuition for what that limit may be. For complicated sequences, this is a difficult task.

What should be somewhat clear, however, is that if a sequence is Cauchy, then it really is the case that the tail of the sequence "gets small and stays small forever." This is not the complication in establishing that a Cauchy sequence converges to a limit; the complication is with *the limit point itself.* In particular, let us not consider a sequence taking values in all of $\mathbb{R}$, but one which we explicitly dictate to take values in a subset of $\mathbb{R}$: if we let

$$a_n := \frac{1}{n},$$

then clearly $a_1 = 1$, $a_2 = 1/2$, $a_3 = 1/3$, ..., and $a_n \to 0$ as $n \to \infty$. However, notice that *none* of the terms $a_n = 0$ exactly; it is only *in the limit* that a_n tends toward $a := 0$. But this means we could very well define the *underlying space* on which the sequence $a_n = 1/n$ is constructed to not be all of $\mathbb{R}$, but just $(0, 1] \subset \mathbb{R}$. Each $a_n = 1/n \in (0, 1]$ for $n \in \mathbb{N}$, so the sequence a_n defined on the space $(0, 1]$ is a completely valid construction. Yet, if we imagine that "the only numbers a we're allowed to name as prospective limits of our sequence a_n are those in $(0, 1]$," it is no longer the case that a_n converges to a limit in a precise sense; since we've explicitly defined the sequence $a_n = 1/n$ to live on the space $(0, 1]$, and since $a_n \to 0 \notin (0, 1]$, what we should perceive as happening is our sequence a_n creeps closer and closer to the edge of the world $(0, 1]$, but we're simply "not able to see where it's heading" — even though we can clearly see each member $a_n = 1/n$ in our space $(0, 1]$, its inevitable destination $a = 0$ remains elusive to us.

This example shows something important: whether or not a Cauchy sequence converges to a limit point is not really a fact about sequences alone — rather, it is a fact about the *space the sequence lives in.* The same list of numbers $1, 1/2, 1/3, ...$ can behave in two *qualitatively* different ways depending on what we "count as a legitimate destination," even if we're able to *quantitatively* ascertain that the destination "should be the same either way." In $\mathbb{R}$, the tail of the sequence $a_n = 1/n$ collapses toward 0, and 0 is a real number, so a_n legitimately converges *in* $\mathbb{R}$. On the other hand, when we restrict our space to just $(0, 1]$, the tail of the sequence $a_n = 1/n$ still collapses, but its natural destination lies just beyond the boundary of description relative to the space we've decided to call our home. That is, $a_n = 1/n$ is always a Cauchy sequence, but a_n does *not* converge to a limit in $(0, 1]$, while it *does* converge to a limit in $\mathbb{R}$.

To summarize, a Cauchy sequence a_n is best thought of as a sequence whose tail eventually agrees on where its going. Past some thresholding index $N \in \mathbb{N}$, all of its terms are packed into an arbitrarily small interval; the sequence has essentially committed to a destination. The important question in deciding whether or not to call such a Cauchy sequence convergent is thus a matter of whether that destination is visible to us in the space we've decided to define the sequence over. Intuitively, it should be clear that sequences defined on a world like $(0, 1]$ tend to be a bit messier than those on $[0, 1]$, simply because we can define sequences like $a_n = 1/n$ whose members all live in $(0, 1]$, but which converge to a point outside $(0, 1]$. This defect is obviously not present when we define $a_n = 1/n$ on $[0, 1]$, since in that case, $a_n \to 0 \in [0, 1]$.

All of this may seem somewhat pedantic, since our quibble with our ostensibly artificial example $a_n = 1/n$ on $(0, 1]$ is essentially a matter of "just one point, $a = 0$." For the most part, this is a valid observation, and it is at the heart of the formal workaround which will allow us to say when Cauchy sequences converge to a limit. However, we quickly want to emphasize the fact that just because this example appears artificial does not mean it is a waste of time to ponder its ramifications. A more interesting question might be whether we could have a sequence of real numbers $a_n \in \mathbb{R}$ which converges to a complex-valued limit, say $a = x + iy \in \mathbb{C}$ with nonzero imaginary part $\mathfrak{I}(a) := y \neq 0$. The reason this may not be immediately obvious is because it certainly is the case that sequences of *rational numbers* $a_n \in \mathbb{Q}$, for instance, can converge to *irrational limits* $a \in \mathbb{R} \setminus \mathbb{Q}$. Consider the sequence

$$1,\ 1.4,\ 1.41,\ 1.414,\ 1.4142, \ldots;$$

each member of this sequence is a rational number (we can represent each by a fraction, say $a_1 = 1/1$, $a_2 = 14/10$, $a_3 = 141/100$, and so on). One could construct such a sequence by However, these are clearly just the first few terms in the decimal expansion of $\sqrt{2}$, and we can imagine continuing the construction in such a way that $a_n \to \sqrt{2}$ as $n \to \infty$. In this case, it is clear that the sequence a_n is Cauchy (why?), and each member $a_n \in \mathbb{Q}$, but by construction $a_n \to \sqrt{2} \notin \mathbb{Q}$.

Motivated by this, you may be tempted to try and construct a sequence of real numbers $a_n \in \mathbb{R}$ whose limit $a \notin \mathbb{R}$; as gestured, one natural starting place would be to check whether real sequences could have a complex limit. However, we quickly see this is impossible; since $\mathbb{R}$ forms a line inside the complex plane, and any non-real complex number lies a nonneglible positive distance above or below that line. Concretely, if $z = x + iy$ with $y \neq 0$, then for every real number r (equivalently, complex number with zero imaginary part), then if we measure the distance from the real r to the complex z by

$$|r - z| = \sqrt{(r - x)^2 + (y)^2} \geq \sqrt{y^2} = |y|.$$

In particular, if we take $\epsilon = |y|/2$, then $|r - z| \geq \epsilon$ for any $r \in \mathbb{R}$. In words, what this means is that the disk of radius $|y|/2$ about the complex point $z = x + iy$ will never contain points on the real-axis. Thus it cannot be that a real sequence a_n converges to a complex limit a.

What we've seen thus far are three examples highlighting different convergence phenomena:

- $\mathbb{Q}$ inside $\mathbb{R}$: we can construct a sequence of rational numbers $a_n \in \mathbb{Q}$ which converge to an irrational limit $\sqrt{2} \notin \mathbb{Q}$.
- $(0, 1]$ inside $[0, 1]$: we can construct a sequence of real numbers $a_n \in (0, 1]$ which converge to an exterior point $0 \notin (0, 1]$.
- $\mathbb{R}$ inside $\mathbb{C}$: any convergent sequence of real numbers $a_n \in \mathbb{R}$ will converge to a real limit, so there is *no* "real-to-nonreal" convergence story to tell.

The three examples above display three different flavors of convergence behavior, and we hasten to discuss each in a bit more detail. Again, it is quite simple to find a sequence of rational numbers converging to an irrational limit; we can construct rational approximations to $\sqrt{2}$ using, say, a bisection or Newton's method applied to $x^2 - 2 = 0$. One can do the same thing for $x^2 - 3 = 0$, $x^2 - 5 = 0$, and so on. The point we seek to make here is that it is intuitively apparent that "irrational numbers exist," even if we're unable to pin them down completely — using a straight-edge to draw the right triangle $a^2 + b^2 = c^2$ with $a = b = 1$ is going to "show you what $\sqrt{2}$ looks like on the hypotenuse $c = \sqrt{2}$," even though we're unable to write the decimal expansion of $\sqrt{2}$. Thus, if we "agree to live in the rational numbers $\mathbb{Q}$," we'll be forced to live with a "relatively hole-y landscape" — this is somewhat different from the second example, where even though $0 \notin (0, 1]$, the notion of "walking across the landscape $(0, 1]$" seems more plausible than "walking across the rational numbers." The last example, then, shows how a relatively *complete* landscape should feel — there is simply no way we can "chart a set of steps across the real-line $\mathbb{R}$" inadvertently leading us *outside* of the real-line $\mathbb{R}$ into, say, the complex numbers $\mathbb{C}$. The subtlety here is that $\mathbb{R}$ can be represented either as its own distinct entity, *or* as a single axis inside the complex plane $\mathbb{C}$, but sequences of real-valued steps a_n will "feel the same" regardless of which world ($\mathbb{R}$ or the real-axis in $\mathbb{C}$) we perceive as truly inhabiting the sequence a_n. This is not the case for rational sequences, where we can "chart sequences of steps leading us outside $\mathbb{Q}$ into $\mathbb{R}$," or for sequences defined on *open subsets* of the real line, where we can "chart sequences of steps leading us to the edge of the open world."

It should hopefully be somewhat apparent that $\mathbb{R}$ occupies something of a dignified air amongst the various number systems: sequences we define over *all of* $\mathbb{R}$ seem to stay in $\mathbb{R}$. But we now take one step further back and ask: what *is* $\mathbb{R}$? It's clear how we can define the rational numbers $\mathbb{Q}$: we can simply take $\mathbb{Q} = \{p/q\colon\ p, q \in \mathbb{Z}\}$. But $\mathbb{R}$ is unwieldy by comparison: it includes numbers like $\sqrt{2}$ (which we can define as the positive root of the equation $x^2 - 2 = 0$) but it also includes numbers like π and e, which are somehow "more complicated to describe" than $\sqrt{2}$. Indeed, we cannot define π to be the solution to an algebraic equation as we can $\sqrt{2}$; the same is true for e.[2] We bring this up merely to notice that the real numbers $\mathbb{R}$ cannot be characterized as just "the set of numbers solving some algebraic equation with rational coefficients" — it is more complicated still, containing a class of numbers that clearly characterize tangible objects of interest in practical applications but which remain elusive to our most familiar algebraic methods.

It is perhaps helpful to consider the number e in a bit more detail. There are several ways to define the number e, but before we remind you of them, consider *why* defining such a number would be of import in the first place. That is, if you were "starting from the beginning" and had no knowledge that exponential growth *should* be characterized by the number e, how could you identify and construct it yourself? Again, we remind you of the distinction between *algebraic* numbers — those which we can identify as the roots of algebraic equations with rational coefficients like $x^2 - 2 = 0$ — and *transcendental* numbers, which at this point are (somewhat vaguely) the "non-algebraic real numbers." But it is somewhat hard to conceive of *why* transcendental numbers should even exist in the first place.

One way to dissolve some of this mystery is to notice that there are really two different notions of "defining a number" floating around in our heads. The algebraic notion is: "give me a finite equation with rational coefficients, and I will point to the real number that solves it." We can identify $\sqrt{2}$, $\sqrt{3}$, $\sqrt{5}$, and many other irrational reals in this manner. However, it is *analysis* which provides us with a second, equally legitimate notion of constructing real numbers: "give me a *process* that produces better and better *rational approximations*, and I will name the destination of that process." In this second sense, a number is not just presented to us as a static object we seek to pin down, but as the *limit of a procedure*.

[2]We call numbers like π and e which are real- (or complex-) valued but which are *not* the solution to an algebraic equation with rational coefficients *transcendental numbers*.

After making this identification, it should be less surprising that $\mathbb{R}$ cannot be pinned down with algebra alone. Indeed, finding roots of a polynomial with rational coefficients is just one kind of rule we can write down, but we know from an introductory exposure that it is the *limit* which is the king of calculus, a fact which can be obfuscated by the time we commit to studying techniques for evaluating derivatives, integrals, and infinite series. But each of these is defined precisely in terms of a limiting procedure: the derivative of a (continuous univariate) function $f(x)$ at a point $x = x_0$ is

$$f'(x_0) = \lim_{h \to 0} \frac{f(x_0 + h) - f(x_0)}{h};$$

the Riemann integral is

$$\int_a^b f(x)\, dx = \lim_{n \to \infty} \sum_{i=1}^{n} f(x_i)\Delta x;$$

and an infinite series is

$$\sum_{i=1}^{\infty} a_i = \lim_{n \to \infty} \sum_{i=1}^{n} a_i.$$

A first course in calculus teaches us how to evaluate these limits in some situations, but we often fail to internalize the fact that the limits themselves are often numbers which otherwise defy description! Perhaps the simplest example to demonstrate this is indeed the number e. If we imagine tracking growth of, say, a bank account, bacterial population, or chemical reaction, then we can often imagine the rate of growth of that population as proportional to the size of the population at the time (a bigger bank account accumulates more interest, a larger bacteria colony produces more offspring, etc.). Concretely, if we have a bank account with $x_0 = 1$ unit at baseline accumulating 100% interest every year, then at the end of the first year we will have accumulated

$$x_1 = x_0 \left(1 + \frac{\%\text{ interest}}{100}\right) = 1\left(1 + \frac{100}{100}\right) = 1(1+1) = 2.$$

However, we could also imagine breaking up the interest deposits into six month increments; say, with 50% interest accumulating after the first six months, and another 50% at the end of the year. Then we would have

$$\begin{aligned} x_1 &= x_0 \left(1 + \frac{50}{100}\right) = 1\left(1 + \frac{1}{2}\right) = 1.5, \\ x_2 &= x_1 \left(1 + \frac{50}{100}\right) = 1.5\left(1 + \frac{1}{2}\right) = 2.25. \end{aligned}$$

Notice that the computation above is done in a recursive manner; in particular, with $x_0 = 1$, we can compactly find x_2 via

$$x_2 = \left(1 + \frac{1}{2}\right)\left(1 + \frac{1}{2}\right) = \left(1 + \frac{1}{2}\right)^2 = 2.25.$$

Similarly, if we start with $x_0 = 1$ and accumulate 25% interest every quarter (three months), then at the end of the year we would have accumulated a total of

$$x_4 = \left(1 + \frac{1}{4}\right)^4 = 2.441406.$$

In general, we imagine that if we start with $x_0 = 1$ and accumulate $100(1/n)\%$ interest every period, then at the conclusion of n periods, we will have accumulated total value

$$x_n = \left(1 + \frac{1}{n}\right)^n. \tag{1.A7}$$

Something we quickly notice if we try plugging in larger and larger values of n int (1.A7) is that it never seems to exceed 3; for example:

```r
xn <- function(n) (1 + 1/n)^n

xn(1) # Annually
```

```
[1] 2
```

```r
xn(2) # Bianually
```

```
[1] 2.25
```

```r
xn(4) # Quarterly
```

```
[1] 2.441406
```

```r
xn(12)
```

```
[1] 2.613035
```

```r
xn(120)
```

```
[1] 2.707041
```

```r
xn(1200)
```

```
[1] 2.71715
```

```r
xn(12000)
```

```
[1] 2.718169
```

```r
xn(120000)
```

```
[1] 2.718271
```

```r
xn(1200000)
```

```
[1] 2.718281
```

Thus, what we observe is that, for instance,

$$\left(1+\frac{1}{12}\right)^{12} \approx 2.613, \quad \left(1+\frac{1}{120}\right)^{120} \approx 2.707, \quad \left(1+\frac{1}{1200}\right)^{1200} \approx 2.717,$$

and by the time n grows quite large, we can observe that the sequence seems to settle down; in particular, it appears as if

$$\left(1+\frac{1}{n}\right)^{n} \overset{n\to\infty}{\longrightarrow} 2.71828...$$

Again, what's important about this is that the number 2.71828... seems to have materialized not from an algebraic equation, but as a result of our defining a *limiting process* and observing where it seems to be headed. The question now is whether we can formally characterize this limiting behavior; i.e., whether it *really is* the case that $x_n := (1+1/n)^n < \infty$ for all $n \in \mathbb{N}$, and if it is, whether we can identify a concrete limiting value (i.e., we must not discount the possibility that x_n behaves in an erratic, unpredictable manner for different magnitudes of $n \in \mathbb{N}$ — it may be the case that x_n *appears* to settle down towards a limit, but that after some threshold a new, possibly unexpected behavior begins to emerge).

Going back to when we plugged in larger and larger values to the sequence $x_n = (1+1/n)^n$, we can observe two critical heuristics in establishing the existence of a limit $\lim x_n$:

(i) **Monotonicity**: It appears as if x_{n+1} is larger than x_n no matter the $n \in \mathbb{N}$ we decide to plug in.

(ii) **Boundedness**: No matter how large an n we plug in to x_n, it appears as if x_n never exceeds, say, the number 3.

Thinking purely from our intuition, a monotone sequence is one which avoids the unpredictability we alluded to earlier; that is, we can expect consistent growth or decay of a monotone sequence as we take larger and larger n, without the possibility of "unexpected turn-arounds." It is not immediately obvious that the sequence under investigation $x_n = (1+1/n)^n$ is monotone; to see this, recall the *binomial formula*, which says that we may expand a binomial $(x+y)^n$ as

$$(a+b)^n = \sum_{k=0}^{n} \binom{n}{k} a^k b^{n-k}.$$

(Notice that we can always flip the positions of a and b in the sum on the left, so that either a or b can be written as powers of k or $n-k$ depending on which is more convenient.) Now, noticing that our sequence is of this form, we can write

$$\left(1+\frac{1}{n}\right)^n = \sum_{k=0}^{n}\binom{n}{k}\left(\frac{1}{n}\right)^k = \binom{n}{0}\left(\frac{1}{n}\right)^0 + \binom{n}{1}\left(\frac{1}{n}\right)^1 + \binom{n}{2}\left(\frac{1}{n}\right)^2 + \cdots + \binom{n}{k}\left(\frac{1}{n}\right)^k + \cdots$$

We can simplify the binomial coefficients as

$$\binom{n}{0}\left(\frac{1}{n}\right)^0 + \binom{n}{1}\left(\frac{1}{n}\right)^1 + \binom{n}{2}\left(\frac{1}{n}\right)^2 + \cdots + \binom{n}{k}\left(\frac{1}{n}\right)^k + \cdots = 1 + n\left(\frac{1}{n}\right) + \frac{n(n-1)}{2!}\frac{1}{n^2} + \cdots + \frac{n(n-1)\cdots(n-k+1)}{k!}$$

and, if we look at only the lattermost general term, we can write

$$\frac{n(n-1)\cdots(n-k+1)}{k!}\frac{1}{n^k} = \frac{1}{k!}\underbrace{\left(\frac{n}{n}\right)}_{=1}\left(\frac{n-1}{n}\right)\cdots\left(\frac{n-k+1}{n}\right) = \frac{1}{k!}\left(1-\frac{1}{n}\right)\left(1-\frac{2}{n}\right)\cdots\left(1-\frac{k-1}{n}\right).$$

Again, this is the kth term in the binomial expansion, so

$$\left(1+\frac{1}{n}\right)^n = \sum_{k=0}^{n}\frac{1}{k!}\left(1-\frac{1}{n}\right)\left(1-\frac{2}{n}\right)\cdots\left(1-\frac{k-1}{n}\right). \tag{1.A8}$$

Now, if we imagine replacing n by $n+1$, what we have is

$$\left(1+\frac{1}{n+1}\right)^{n+1} = \sum_{k=0}^{n+1}\frac{1}{k!}\left(1-\frac{1}{n+1}\right)\left(1-\frac{2}{n+1}\right)\cdots\left(1-\frac{k-1}{n+1}\right) \tag{1.A9}$$

Notice that each (1.A8) and (1.A9) are the product of numbers of the form (1−small positive number) $<$ 1, but that the terms in (1.A8) are each slightly smaller than their corresponding terms in (1.A9) since (for example)

$$\frac{1}{n} > \frac{1}{n+1} \implies 1-\frac{1}{n} < 1-\frac{1}{n+1}.$$

Each of the $k = 0, 1, \ldots, n$ terms in (1.A9) will thus be slightly larger than the corresponding terms in (1.A8). Furthermore, there is an *additional* positive $n+1$ term in (1.A9) not present in (1.A8); it is thus clear that (1.A9) sums more terms, each of which is slightly larger than the corresponding terms in (1.A8) — we thus conclude that, indeed,

$$\text{For all } n \in \mathbb{N}\colon\ x_{n+1} = \left(1+\frac{1}{n+1}\right)^{n+1} > \left(1+\frac{1}{n}\right)^n = x_n.$$

We've just shown that the sequence $x_n = (1+1/n)^n$ is *monotone increasing*; its terms $x_{n+1} > x_n$ for all $n \in \mathbb{N}$. There are really two possibilities: either $x_n \to \infty$ as $n \to \infty$, or $x_n \to x < \infty$ for some number x. From our earlier inspection, it already seems likely that x_n converges, since on plugging in larger and larger n we were unable to find an x_n exceeding the number 3. However, to be rigorous, we can pursue an argument in a manner similar to the above: we already observed in (1.A8) that

$$\left(1+\frac{1}{n}\right)^n = \sum_{k=0}^{n}\frac{1}{k!}\left(1-\frac{1}{n}\right)\left(1-\frac{2}{n}\right)\cdots\left(1-\frac{k-1}{n}\right),$$

where each term in the sum is comprised of products of numbers strictly less than 1. That is,

$$\frac{1}{k!}\left(1-\frac{1}{n}\right)\left(1-\frac{2}{n}\right)\cdots\left(1-\frac{k-1}{n}\right) < \frac{1}{k!},$$

since the left hand side is $1/k!$ multiplied by a number less than 1. Since this inequality holds for each term in the sum, it will also hold over the sum itself; in particular,

$$x_n = \left(1+\frac{1}{n}\right)^n = \sum_{k=0}^{n}\frac{1}{k!}\left(1-\frac{1}{n}\right)\left(1-\frac{2}{n}\right)\cdots\left(1-\frac{k-1}{n}\right) < \sum_{k=0}^{n}\frac{1}{k!}.$$

Now, from

$$x_n < \sum_{k=0}^{n} \frac{1}{k!} = 1 + 1 + \frac{1}{2!} + \cdots + \frac{1}{n!},$$

observe that for each $k \geq 2$,

$$k! = k(k-1)\cdots(2)(1) \geq 2(2)\cdots(2)(1) = 2^{k-1},$$

where we're just "replacing the larger integers k, $k-1$, ... on the left with 2s on the right." This means that

$$x_n < \sum_{k=0}^{n} \frac{1}{k!} \leq 1 + \left(\frac{1}{2^0} + \frac{1}{2^1} + \frac{1}{2^2} + \cdots + \frac{1}{2^{n-1}}\right) = 1 + \sum_{k=0}^{n-1} \frac{1}{2^k}.$$

To summarize, the reason we've argued in this manner is precisely because we don't have a *closed-form expression* for evaluating the sums we've used to bound our original binomial expansion,

$$\left(1 + \frac{1}{n}\right)^n = \sum_{k=0}^{n} \binom{n}{k} \left(\frac{1}{n}\right)^k.$$

Binomial coefficients are tough to work with in this form, so we expanded the sum as

$$\left(1 + \frac{1}{n}\right)^n = \sum_{k=0}^{n} \frac{1}{k!} \left(1 - \frac{1}{n}\right) \left(1 - \frac{2}{n}\right) \cdots \left(1 - \frac{k-1}{n}\right).$$

Still, we don't have a clean way to evaluate the sum on the right without computing all n terms in the sum individually and combining them at the end — if our goal is to show that the sum is bounded for *all* $n \in \mathbb{N}$, the form above will be insufficient. However, we don't need to evaluate the sum itself to show it is bounded; if we can compare it to a sum which is strictly larger for any $n \in \mathbb{N}$ and show that *that* sum is bounded, the boundedness of the original will follow by monotonicity. (Notice that monotonicity is necessary for this argument to work as described; otherwise, it could be that the original sum is strictly less than our comparator for all $n \in \mathbb{N}$, but that the original reverses course and begins to decrease after some n.)

In pursuit of this, we established that

$$\left(1 + \frac{1}{n}\right)^n < \sum_{k=0}^{n} \frac{1}{k!},$$

which certainly appears to be a simpler sum to work with. The last maneuver is perhaps the most surprising one; rather than working directly with the sum above, we instead bounded it by

$$\sum_{k=0}^{n} \frac{1}{k!} \equiv 1 + \sum_{k=1}^{n} \frac{1}{k!} < 1 + \sum_{k=0}^{n-1} \frac{1}{2^k}.$$

Now, it may not be entirely evident why bounding $\sum_k 1/k!$ by $\sum_k 1/2^k$ is in any way helpful, but let us realize *why* the former is "difficult," and why the latter is somewhat "easy" by comparison. Define

$$S_n := \sum_{k=0}^{n} \frac{1}{2^k} \equiv 1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \cdots + \frac{1}{2^n}.$$

If we peel off the first $k = 0$ term from S_n, what we're left with is

$$\frac{1}{2} + \frac{1}{4} + \cdots + \frac{1}{2^n} = \frac{1}{2}\left(1 + \frac{1}{2} + \frac{1}{4} + \cdots + \frac{1}{2^{n-1}}\right) = \frac{1}{2}S_{n-1},$$

so that, in particular, we can establish the recursive relationship between S_n and S_{n-1} for any $n \in \mathbb{N}$:

$$S_n = \frac{1}{2}S_{n-1} + 1. \tag{1.A10}$$

Realize, however, that *in the limit* as $n \to \infty$, both S_n and S_{n-1} describe the same infinite series — namely,

$$\lim_{n\to\infty} S_n = \lim_{n\to\infty} S_{n-1} = \sum_{k=0}^{\infty} \frac{1}{2^k}. \tag{1.A11}$$

This is quite an important realization. Defining

$$S := \lim_{n\to\infty} S_n = \sum_{k=0}^{\infty} \frac{1}{2^k} \tag{1.A12}$$

and recalling that for a convergent sequence $x_n \to x$ as $n \to \infty$ and real constants $a, b \in \mathbb{R}$, we have

$$\lim_{n\to\infty} ax_n + b = a(\lim_{n\to\infty} x_n) + b = ax + b,$$

what (1.A10) and (1.A11) imply is that

$$\lim_{n\to\infty} S_n = \frac{1}{2}(\lim_{n\to\infty} S_{n-1}) + 1,$$

or

$$S = \frac{1}{2}S + 1 \implies S = \frac{1}{1 - \frac{1}{2}} = 2.$$

In particular, notice that the first term in the sum (1.A12) is $a = 1$, and that each successive term differs from the previous by a *common ratio* of $r = 1/2$. We thus might think it sensible to, in general, claim that

$$S = a + rS \implies S = \frac{a}{1-r}.$$

The natural question is for which values of $r \in \mathbb{R}$ and $a \in \mathbb{R}$ this holds true — we claim it holds for $a = 1$ and $r = 1/2$, but is it always the case?

We take this opportunity to present a general definition.

Definition 1.0.3 (Geometric Series)

Let

$$S_n := a + ar + ar^2 + \cdots + ar^n = \sum_{k=0}^{n} ar^k.$$

Then, for any ratio $r \in \mathbb{R}$ and initial point $a \in \mathbb{R}$, the value of S_n for a given $n \in \mathbb{N}$ is

$$S_n := \sum_{k=0}^{n} ar^k = \frac{a(1-r^{n+1})}{1-r}.$$

Furthermore, suppose that $-1 < r < 1$ strictly. Then

$$S = \lim_{n\to\infty} S_n \quad \text{exists,}$$

and

$$S = \lim_{n\to\infty} S_n = \lim_{n\to\infty} \sum_{k=0}^{n} ar^k := \sum_{k=0}^{\infty} ar^k = \frac{a}{1-r}; \quad -1 < r < 1, \quad a \in \mathbb{R}. \qquad \text{(Geometric Series)}$$

Remark.

Notice that the *partial sum* of the geometric series is always well-defined (i.e., for any ratio $r \in \mathbb{R}$) — its value is

$$S_n = \sum_{k=0}^{n} ar^k = \frac{a(1-r^{n+1})}{1-r}, \quad \text{For all } r \in \mathbb{R}.$$

On the other hand, we only define the *limit* for $|r| < 1$ — i.e.,

$$\lim_{n\to\infty} S_n = \sum_{k=0}^{\infty} ar^k = \frac{a}{1-r} \iff -1 < r < 1.$$

The reason for this is readily apparent when we take the limit relative to the closed-form expression for the partial sum. What we have is

$$\lim_{n\to\infty} S_n = \lim_{n\to\infty} \frac{a(1-r^{n+1})}{1-r} = \frac{a(1-\lim_{n\to\infty} r^{n+1})}{1-r}.$$

One then observes that

$$\lim_{n\to\infty} r^{n+1} = \begin{cases} 0, & -1 < r < 1, \\ \infty, & r \geq 1, \\ \text{DNE}, & r \leq -1. \end{cases}$$

We say that the limit is ∞ in the case where $r > 1$, but that the limit *does not exist* whenever $r < -1$. The reason for this is that when $r > 1$, $r^n < r^{n+1}$ for any $n \in \mathbb{N}$; on the other hand, when $r < -1$, the powers r^n differ in sign depending on whether n is even or odd! It is thus the case that r^n is monotonic but unbounded for $r > 1$, but is neither monotonic nor bounded for $r < -1$. Since it would seem we "know a bit more about where the sequence is headed" in the former case, it is common to distinguish these cases by using the symbols $\pm\infty$ whenever a sequence is monotonic but unbounded, and to use DNE whenever some sort of *periodic* behavior is displayed by a sequence. (We'll formalize this notion using the $\limsup$ and $\liminf$ in the coming sections.)

The example we've just gone through illustrates the evaluation of a geometric series with $a = 1$ and $r = 1/2$ — that is,

$$S = \sum_{k=0}^{\infty} \frac{1}{2^k} = \frac{a}{1-r} = \frac{1}{1-\frac{1}{2}} = 2.$$

What this means in the context of our current derivation is that

$$\left(1 + \frac{1}{n}\right)^n < 1 + \sum_{k=0}^{n-1} \frac{1}{2^k},$$

so that, in the limit as $n \to \infty$,

$$\lim_{n\to\infty} \left(1 + \frac{1}{n}\right)^n < 1 + \lim_{n\to\infty} \sum_{k=0}^{n-1} \frac{1}{2^k} = 1 + \lim_{n\to\infty} S_{n-1} = 1 + S = 1 + 2 = 3.$$

What we've just shown is that, for the sequence

$$x_n = \left(1 + \frac{1}{n}\right)^n,$$

we have both

$$x_n < x_{n+1}, \quad \text{For all } n \in \mathbb{N}, \qquad (x_n \text{ is Monotone Increasing})$$

and

$$\lim_{n\to\infty} x_n < 3. \qquad (x_n \text{ is Bounded})$$

Again, what we're trying to do here is to evaluate $\lim_{n\to\infty}(1 + 1/n)^n$. We used the binomial formula to write this as an infinite series, but since evaluating infinite series is quite difficult, we wanted to check whether it even makes sense that the series would converge to a finite value as $n \to \infty$. Having seemed to confirm this (or at least having not shown the contrary to be obvious), what we'd like to do ideally is find the actual value of this limit — i.e., to find the value of the limit. Let us agree to call this limit e — i.e., we *define* the number e to be

$$e := \lim_{n\to\infty} x_n;$$

that is, we define

$$e := \lim_{n\to\infty} \left(1 + \frac{1}{n}\right)^n = \lim_{n\to\infty} \sum_{k=0}^{n} \frac{1}{k!} :\equiv \sum_{k=0}^{\infty} \frac{1}{k!}. \qquad \text{(Definition of } e \text{ (i))}$$

1.1 Euclidean Constructions

At this point, we urge you to cast aside any pre-existing notions of "what the number e is supposed to be" — you almost certainly know it to be $e \approx 2.71828\ldots$, but this is not the point. It is our observation that, often, it feels as though numbers like e are "handed from on high"; as though the decimal 2.71828 is the "object of significance." But again, we just defined e to be the *limit of a process* — that this limit "happens to admit a decimal expansion of 2.71828" is better understood as a useful coincidence rather than *the fundamental object* that "e is."

The reason we hasten to make the distinction is because a number's decimal representation seemingly falls into one of two camps — it is relatively common knowledge that a rational number's decimal expansion terminates in some sort of identifiable pattern, while an irrational number's does not.

Yet we can't help but feel we've been somewhat cavalier in tossing around symbols like $\mathbb{Q}$, as if you certainly know what they mean. Some might feel it coddling, but we if I asked you to articulate the difference *precisely*, what would you say?

You might, for example, *define* a **rational number** q to be one admitting a **representation** as a ratio of two integers, say

$$q = \frac{a}{b}, \qquad a, b \in \mathbb{Z} \text{ with } b \neq 0.$$

Here, we use the symbol $\mathbb{Z}$ to denote the set of integers:

$$\mathbb{Z} = \{\ldots, -2, -1, 0, 1, 2, \ldots\},$$

and we stipulate that the denominator $b \neq 0$ of the rational $q = a/b$ be nonzero for precisely the "usual reason": that is, division by zero is an *undefined operation.* But this too feels cavalier. If someone asked you to explain *why* division by zero is undefined — not as a *rule*, but as a *fact* — what would you say?

After some thought, you'd probably arrive at something like: " 'to divide a by b' is to ask 'how many copies of b can fit inside a?' If $b = 0$, the question becomes 'how many copies of 0 fit inside a?' — and for any $a \neq 0$, the answer is that '*no finite number of zeros will ever accumulate to a.*' " What we'd emphasize, then, is that the "operation of division by zero" *does not* fail because of some "arbitrary rule"; it fails because the *question has no coherent answer.*

But notice where we went to obtain our explanation. To explain a/b, the division of a by b, as "how many copies of b fit inside a," we *do not* reach limits, or series, or anything else in the analytic toolkit that seemed to have served us well in the first section. Instead, we said something *physical*: "fitting copies of one thing inside another."

This is one of the most devastating misconceptions in *all* of modern mathematics — to believe that, at a point, we "abandon these rote physical intuitions in favor of something *truer*;" that "such an elementary perspective on divison *must be erroneous* because it is *simple to understand.*"

Here, we dispel such notions entirely. The explanation we just gave for division is not a "pedagogical simplification of something more sophisticated" — it *is* the sophisticated thing. It is the picture in the back of every working mathematician's mind when they are *actually dividing.* The number theorist who "asserts $\gcd(12, 3) = 3$" has concluded *precisely* the same fact the third grader who says "a whole number of 3s fit in 12." Neither is doing a simplified version of what the other does. And to believe otherwise — to imagine that somewhere beyond the elementary explanation lies a "truer" one, accessible only to those who have "earned it" through sufficient abstraction — is to confuse the *language* in which we express an idea with the *idea itself.*

Nevertheless, there are *certainly compelling reasons* for abstraction. If we *only* think of division as “fitting copies,” we can answer the question “$12 ÷3 = ? $” with no difficulty — exactly 4 copies of 3 fit into 12, with nothing left over. Both the number theorist and the third grader similarly conclude

$$12 \div 3 = 4,$$

and that is that.

But we have a harder time with, say, “$12 \div 5 = ?$,” where two copies of 5 fit inside 12, but leave a **remainder** of 2. To do this physically, we might imagine we have a stick of length 12 and a pile of sticks of length 5: we start by noting that we can fit two of the length-5 sticks into the length-12 stick, but since the two length-5 copies together only span 10 out of the 12 lengths, there are 2 units of the length-12 stick left uncovered.”

What do we do with the leftover piece of stick? One option is to simply declare that $12 \div 5$ equals 2 with remainder 2 — perhaps we’d write this succinctly as

$$12 \div 5 = 2\text{R}2.$$

This *is a* correct answer. And if our intent was to, say, reconstruct the length-12 stick as closely as possible *only* out of length-5 sticks, then the answer “$12 \div 5 = 2\text{R}2$” is seemingly the best we could do.

But this is not the only question worth asking. We have a leftover piece of length 2. Rather than discarding it, we can *use* it: “does this piece fit evenly into the 5-stick?” We check, find that two copies of 2 span a stick of length-4, leaving a *new* remainder — a stick of length 1. We record this as

$$5 \div 2 = 2\text{R}1,$$

Again we can repeat the process: the length-2 stick fails to reconstruct the length-5 stick exactly, leaving a leftover bit of length-1 stick. Is there a way to reconstruct the length-2 stick by the leftover of length-1? This time, we place two length-1 sticks and observe they *perfectly recreate* the stick of length-2. We record

$$2 \div 1 = 2\text{R}0.$$

Remark.

And it is worth emphasizing here that, in writing $2 \div 1 = 2\text{R}0$, we’ve explicitly identified that the division terminates with a ”remainder of zero.” Contrast this with what we’d written before:

$$12 \div 3 = 4.$$

Fundamentally, both $2 \div 1$ and $12 \div 3$ conclude the same way, so why hadn’t we *started* with the remainder notation and written

$$12 \div 3 = 4\text{R}0?$$

The answer is that, at the time, we simply hadn’t a need for such specificity. The notion of ”zero remainder” is, philosophically, only meaningful when contrasted by a ”positive remainder.” This is something of a recurring theme in mathematics, and when it gets bewildering is when we ”read the more sophisticated notation” without having taken the time to *appreciate* why such sophistication may be warranted. Indeed, there is essentially always a kernel of intuition underlying even the most sophisticated mathematical processes — something ”you could have thought of yourself,” if you'd been prodded in just the right way to do so. And if someone *ever* tries to convince you otherwise — that their work is of such sophistication that it would be a waste of their time to *even try* and explain its intuitive stature — we contend it is *they* who are truly unsophisticated; *not you.*

But now, let us take a step back and look at what we've *gotten* out of "successively dividing the remainders." We didn't just divide 12 by 5; we performed a *chain* of divisions, each one feeding its remainder in as the next round's divisor:

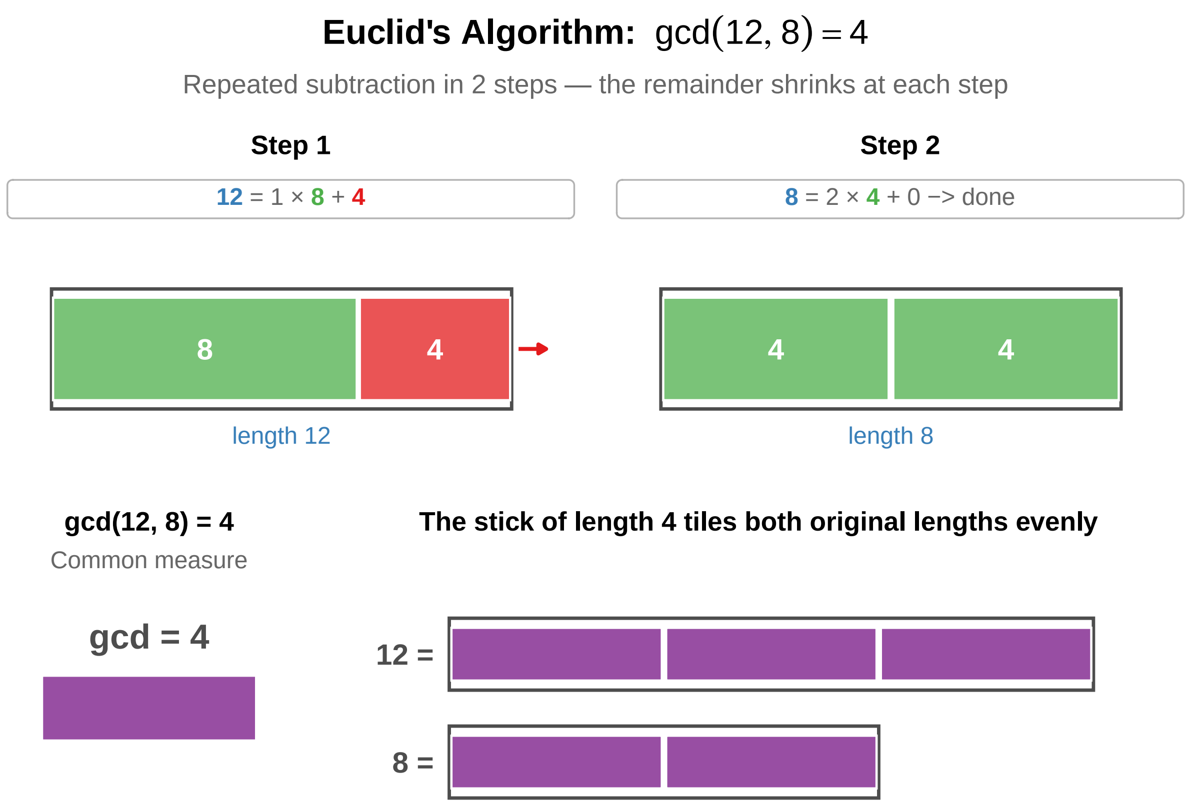
$$\begin{aligned} 12 &= 2 \times 5 + 2, \\ 5 &= 2 \times 2 + 1, \\ 2 &= 2 \times 1 + 0. \end{aligned}$$

At each step, the remainder shrank: 2, then 1, then 0. The process terminated not because we *chose* to stop, but because there was nothing left to divide. And the last nonzero piece — the stick of length 1 — is the one that finally fit evenly. But notice: if the 1-stick tiles the 2-stick, and the 2-stick was the remainder from dividing the 5-stick, and the 5-stick was the remainder from dividing the 12-stick, then the 1-stick must tile *all of them.* (Five copies reconstruct the 5-stick; twelve copies reconstruct the 12-stick.) The chain of divisions has not merely computed a quotient — it has identified a **common measure** of 12 and 5: a single length that reconstructs both originals exactly.

Admittedly, a common measure of 1 is rather boring — a unit stick tiles *any* integer length, by definition. The more interesting case is when the common measure is larger. Consider the pair 12 and 8:

$$\begin{aligned} 12 &= 1 \times 8 + 4, \\ 8 &= 2 \times 4 + 0. \end{aligned}$$

Two steps. The last nonzero remainder is 4, and indeed: three copies of 4 tile 12, while two copies tile 8. The stick of length 4 is not just *a* common measure of 12 and 8 — it is the *largest* such measure, the **greatest common divisor**, written $\gcd(12, 8) = 4$.



What we have just performed — "divide, take the remainder, use it as the next divisor, repeat until the remainder vanishes" — is an *algorithm*: a deterministic, terminating procedure that produces the greatest common divisor of any two positive integers as its output. For integers, termination is guaranteed: the remainders form a strictly decreasing sequence of nonnegative integers, and no such sequence can decrease forever.

The algorithm appears in Book VII of Euclid's *Elements*, composed around 300 BCE, though the method was almost certainly known to earlier Greek mathematicians. It is, as far as we are aware, the oldest algorithm that survives in the written mathematical record. Euclid would not have written $12 = 1 \times 8 + 4$; he would have laid two rods side by side and physically compared them. But the logic is identical to ours: the shorter length is laid along the longer, the excess is noted, the roles reverse, and the process continues all the way until the remainder vanishes. The last nonzero remainder is the common measure.

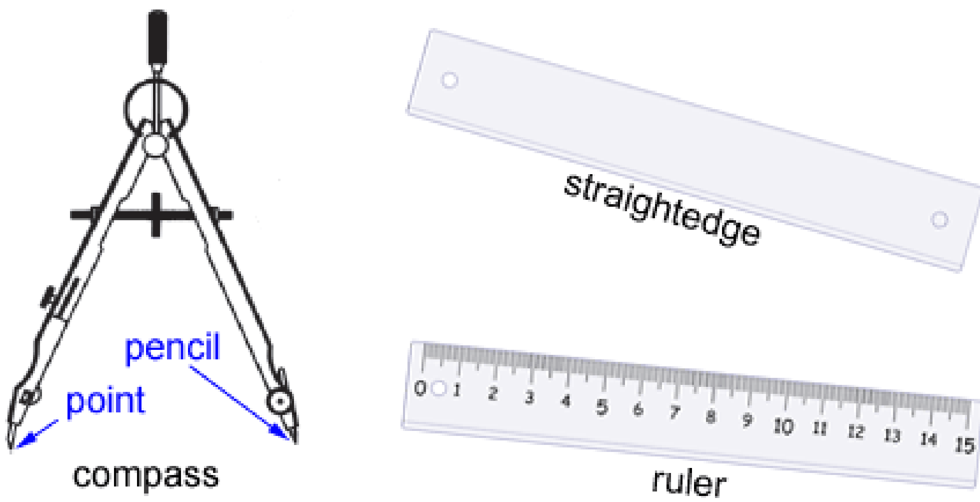
We emphasize the word *until*. For positive integers, the algorithm always terminates, and the common measure always exists. But Euclid was not only interested in integers. He was a *geometer*, and a geometer's magnitudes are not constrained to be whole numbers. The diagonal of a unit square is a perfectly good length — one can hold it, see it, construct it with a compass and straightedge. The question Euclid would have asked is not "is $\sqrt{2}$ rational?" — a question that requires the concept of $\mathbb{Q}$, which he did not possess — but rather: *do the diagonal and the side of a square admit a common measure?* Is there a shorter stick, however small, that tiles both?

This is the question we now pursue. And it is here that the algorithm reveals something remarkable: when applied to two lengths that do *not* admit a common measure, the algorithm does not terminate — but the manner in which it fails to terminate is not chaos. It is structured, periodic, and deeply informative. The failure mode is the information.

Perhaps more concretely, suppose you were asked to draw the number $\sqrt{2}$ on the usual 2D Euclidean plane; how would you do it? Well, because $\sqrt{2}$ is the unique positive root to the algebraic equation $x^2 = 2$, we can use the Pythagorean formula: the right triangle whose opposite and adjacent sides are of unit length $a = b = 1$ has hypotenuse of length c satisfying

$$a^2 + b^2 = c^2 \implies c^2 = 1^2 + 1^2 = 2 \implies c = \sqrt{2}.$$

In this way, we can imagine that $\sqrt{2}$ is somewhat tangible in spite of its irrationality — given only a *straightedge ruler* and a *compass* (see the figure below), we're able to construct the length $\sqrt{2}$ in exactly three physical moves (draw $a = 1$, a perpendicular $b = 1$, and connect the ends). In general, we can pursue an algorithm like the following to construct a length $\sqrt{x}$ for some given length $x > 0$:



Straightedge-and-Compass (https://www.math.net/geometric-construction)

(i) Assume we start with a unit segment of length 1 and the positive length $x > 0$. On a line, mark three points A, B, C such that $AB = 1$ and $BC = x$; hence $AC = 1 + x$.

(ii) Using the compass, draw the semicircle with diameter AC.

(iii) Draw the perpendicular through B and let D be the point this perpendicular intersects the semicircle.

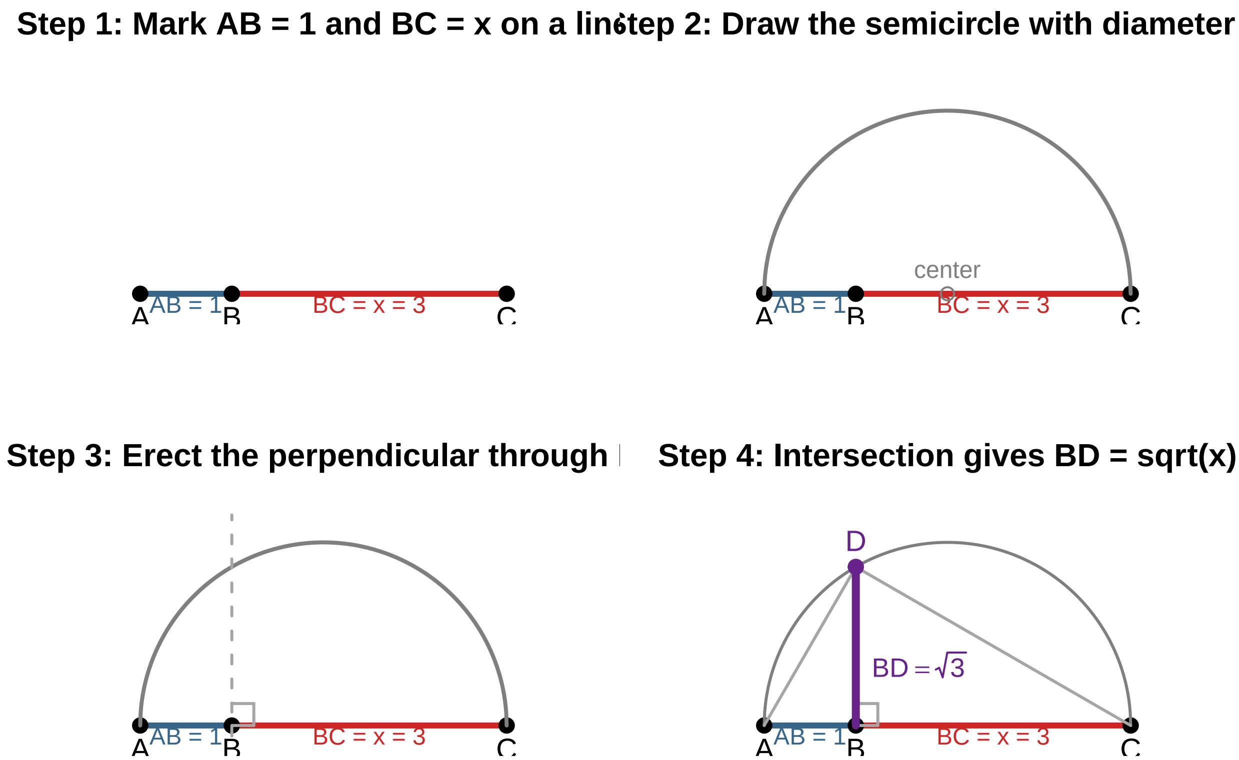
(iv) Then, the new segment BD is of length $\sqrt{x}$.

This algorithm arises simply from the self-similarity of the triangles constructed in this manner. In particular, $\triangle ADB$ is a right triangle, and the *altitude* BD splits this big right triangle into two smaller right triangles $\triangle ABD$ and $\triangle CBD$, each similar to the original. A fact about similar triangles is that their corresponding sides must satisfy the proportionality relations

$$BD^2 = AB \cdot BC;$$

whence, if $AB = 1$ and $BC = x$, then $BD^2 = 1 \cdot x$ and

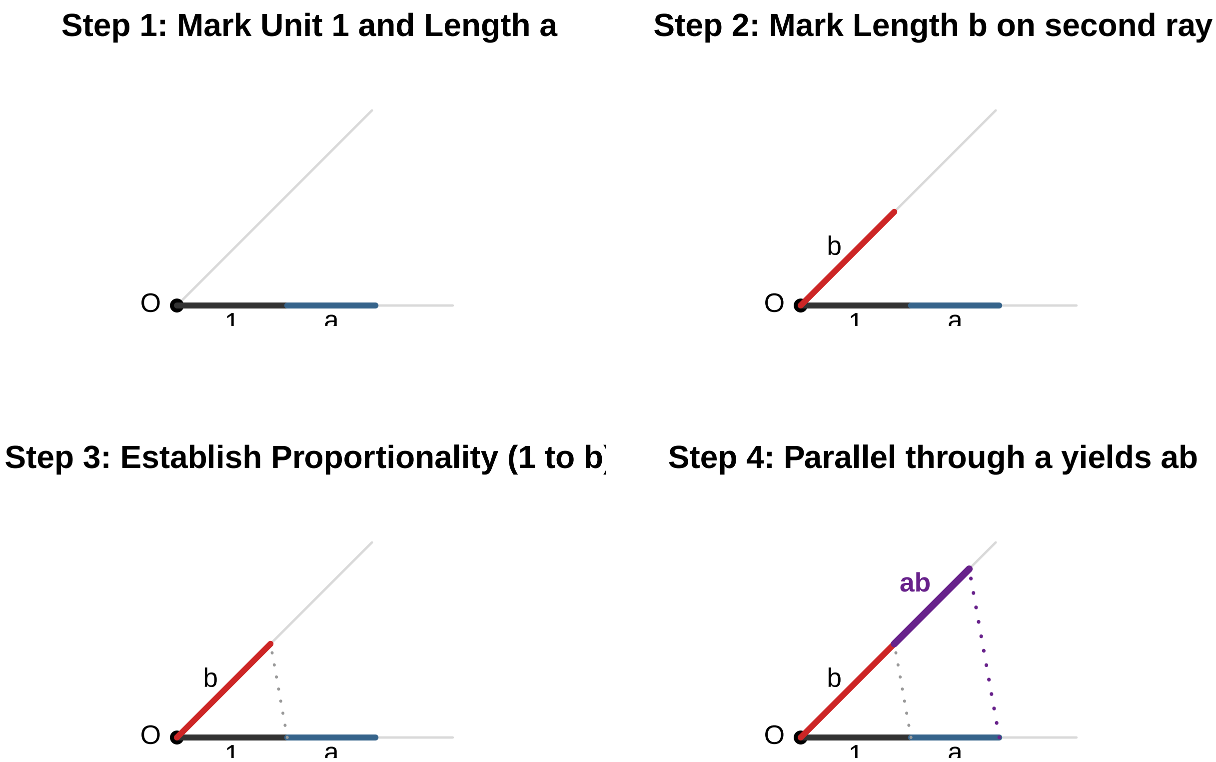
$$BD = \sqrt{1 \cdot x} = \sqrt{x}.$$



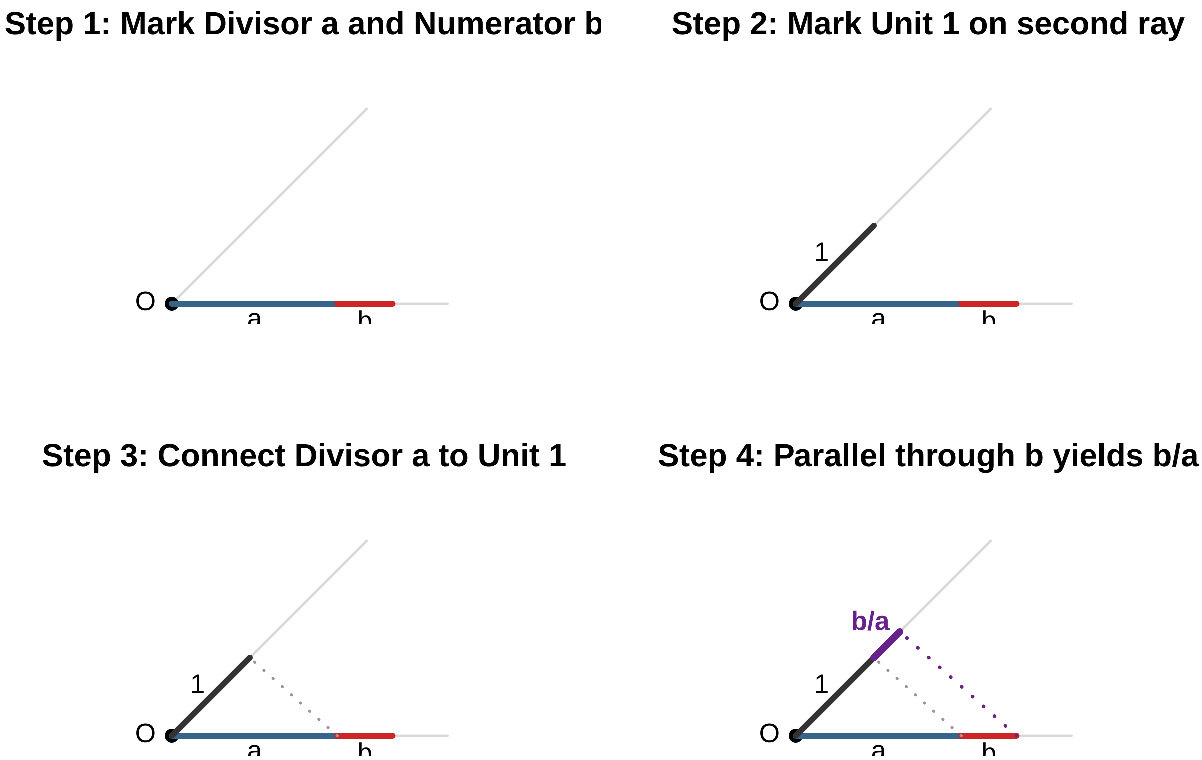
Thus, we've illustrated a concrete, physical manner in which we can construct a segment of length $\sqrt{x}$ for any positive $x > 0$ in the usual 2D Euclidean plane. The point, though, isn't really to convince you that "the number $\sqrt{2}$ exists" — this is already intuitively obvious *because* we can describe $\sqrt{2}$ algebraically as the unique positive root to the equation $x^2 = 2$ over the real line $\mathbb{R}$. The point is moreso that we can actually construct a representation for the *physical magnitude* of $\sqrt{2}$ through a *finite process* of construction. In some way, this makes the existence of $\sqrt{2}$ feel grounded in the same Euclidean reality as the number 1.

This tangibility extends beyond the extraction of roots. If we accept the existence of a unit length 1, we can effectively turn the plane into something of an arithmetic engine. Concretely, realize that though we hadn't formally defined the concept in a Euclidean sense, the construction above effectively describes an intuitive means for *multiplication*: if $AB = 1$ is of unit length and we use our straightedge and compass to draw a point C a distance of x units to the right of B so that $BC = x$, then by drawing the semicircle connecting the endpoints A and C, the point D where the perpendicular erected through AB at the point B hits that semicircle is of length $\sqrt{1 \cdot x}$, or $\sqrt{AB \cdot BC}$.

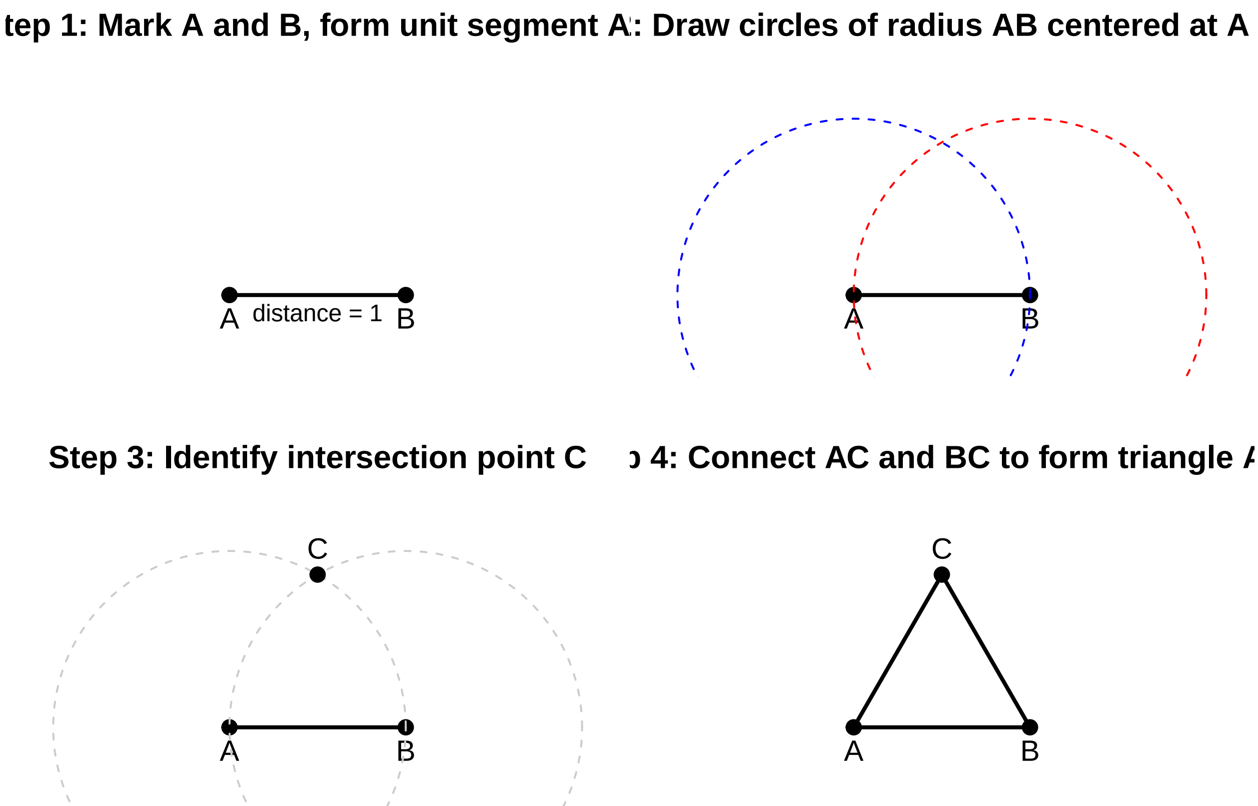
To multiply line segment a by segment b, we first construct a reference triangle using our unit length. We lay out the unit 1 and the length a along our horizontal ray, and mark the length b along a second, angled ray. By connecting the unit point 1 to the point b on the second ray, and then constructing the *parallel* to that line passing through a, the new point of intersection on the ray defined by extending b in the same direction will be of length ab:



The process for division is the precise inverse of multiplication. If we wish to find b/a, we begin with the divisor a and the numerator b on the primary ray, and mark our unit 1 on the second. By connecting the divisor a back to the unit 1, then constructing the parallel passing through the end of the segment b, the intersection with the ray defined by extending the unit length 1 will be of length b/a:



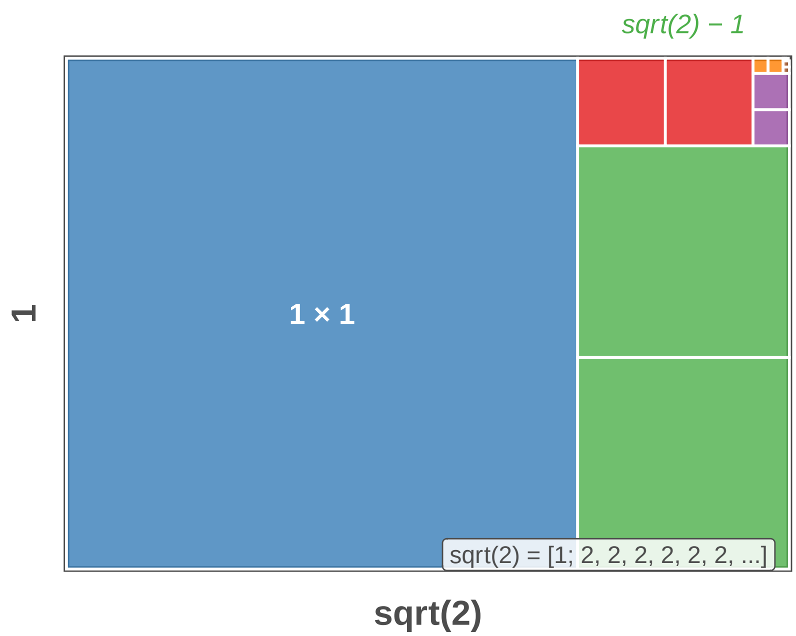
What these constructions reveal is that arithmetic is not a magical operation that happens "to" numbers; it is a *scaling transformation*. By anchoring our triangles at the origin, we see that the segment ab is simply the segment b stretched by the factor a. This realization provides something of a useful reassurance: if we can multiply, divide, and take square roots using nothing but a straightedge and compass, then the entire field of *constructible numbers* — the numbers reachable through finite algebraic steps — should be tangibly accessible to us in our physical reality.

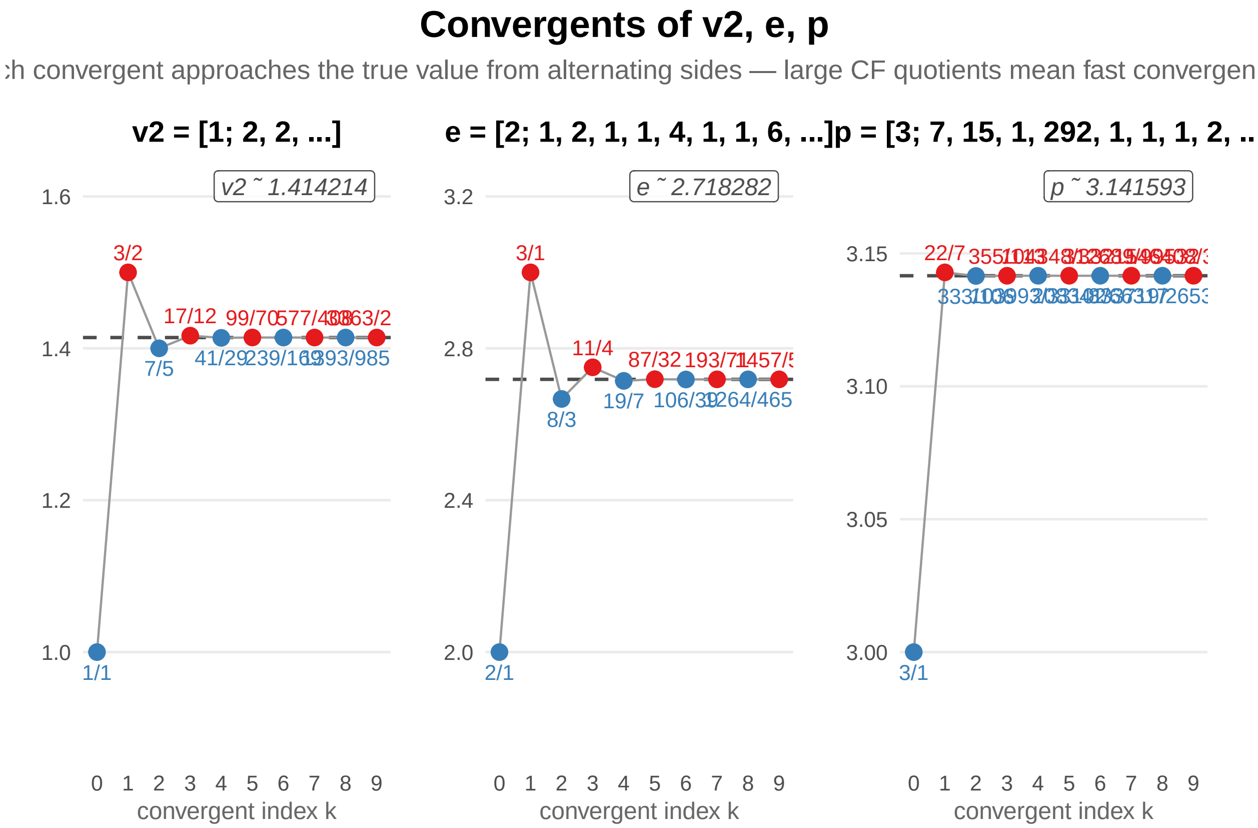


tep 1: Mark A and B, form unit segment A
: Draw circles of radius AB centered at A
A
distance = 1
B
A
B
Step 3: Identify intersection point C
C
A
B
› 4: Connect AC and BC to form triangle /
C
A
B

The self−similar rectangle: sqrt(2) is irrational

Each remainder rectangle has the same aspect ratio as the original — the algorithm never terminates




Convergents of v2, e, p
:h convergent approaches the true value from alternating sides — large CF quotients mean fast convergen
v2 = [1; 2, 2, ...]
e = [2; 1, 2, 1, 1, 4, 1, 1, 6, ...]
p = [3; 7, 15, 1, 292, 1, 1, 1, 2, ..
v2 ~ 1.414214
e ~ 2.718282
p ~ 3.141593
1/1
3/2
7/5
17/12
41/29
99/70
2/1
3/1
8/3
11/4
19/7
87/32
22/7
convergent index k
convergent index k
convergent index k